SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – SHORT ANSWER TEST

Course Code:	CSA16	Course Title:	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL1, BL2)	Assessment:	Assessment Tool 1 – Short Answer Test
Weightage:	10% of CIA	Total Marks:	100 (5 Questions × 20 Marks)
CO1 Statement:	*Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2*		
Student Name:	J. Ramesh	Reg. No.:	192311230
Section / Batch:	CSE	Date:	17/07/2026

Instructions to Students

- Answer all questions. Each question carries 20 Marks. Total: 100 Marks.
- Clearly show all requirements, process, operations.
- Use appropriate models, schema represented scenario wise
- Justify each step in different dimension specified and measures clearly.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Short Answer Test	Architecture of Data Warehouse, ETL, OLAP	Q1 – Q5	Q1 –20 Q2 –20 Q3 –20 Q4 –20 Q5 –20
		GRAND TOTAL:	100 Marks

SECTION A – SHORT ANSWER TEST

1. Suppose that a data warehouse for City Hospital consists of the following four dimensions: patient, doctor, department, and time, and two measures visit_count and avg_charge. At the lowest conceptual level, avg_charge stores the actual consultation fee charged for a patient visit. At higher conceptual levels, it stores the average fee.

(a) Draw a snowflake schema diagram for the data warehouse.

(b) Starting with the base cuboid [patient, doctor, department, time], what specific OLAP operations should be performed to list the average consultation fee department-wise for each year?

(c) If each dimension has four levels (including all), how many cuboids will this cube contain (including base and apex cuboids)?

2. A smart city project is collecting data from multiple sources, including:

- Traffic sensors for vehicle counts and congestion levels.
- IoT devices for monitoring air quality, noise levels, and energy consumption.
- Public transportation systems for ridership data and punctuality.
- Citizen feedback from a mobile app.

Questions:

Fact and Dimension Design:

- Design a star schema to analyze traffic congestion patterns, energy consumption trends, and public transportation efficiency.
- What measures and dimensions would you include? Justify your design.

3. Suppose that a banking data warehouse contains two fact tables: Transaction_Fact and Loan_Fact, sharing dimensions customer, branch, time, and account_type.

- Transaction_Fact → transaction_count, amount
- Loan_Fact → loan_amount, interest

(a) Draw a galaxy schema diagram.
(b) Identify OLAP operations to obtain branch-wise average transaction amount and loan amount yearly.
(c) If each dimension has four levels, calculate total cuboids.

4. A manufacturing company monitors product quality and production efficiency. The data warehouse contains:

- Dimensions:
  - Product: Category, type, and batch.
  - Factory: Location, department, and line number.
  - Time: Year, quarter, month, and week.
  - Supplier: Region, rating, and contract type.
- Measures:
  - Defect rate.
  - Production units.
  - Supplier reliability percentage.

Questions:

- Roll-up: Summarize defect rates by product category for the current year across all factories.
- Drill-down: Focus on a specific factory to analyze defect rates by department and product type.
- Slice: Isolate production unit data for suppliers with a rating of 4+ across all regions.
- Dice: Analyze defect rates and supplier reliability for selected product categories and factory locations.

5. A retail chain manages inventory and sales data across multiple stores. The data warehouse contains:

- Dimensions:
  - Product: Category, subcategory, and SKU.
  - Store: Region, city, and store ID.
  - Time: Year, quarter, month, and day.
  - Customer: Membership tier, age group, and gender.
- Measures:
  - Inventory turnover.
  - Total sales revenue.
  - Units sold.

Questions:

1. Roll-up: Summarize inventory turnover by product category across all stores for the last quarter.
2. Drill-down: Focus on sales data for electronics in a specific region, analysing by subcategory and SKU.
3. Slice: Isolate sales revenue data for male customers in the Bronze membership tier across all stores.
4. Dice: Analyze sales revenue and units sold for selected cities and product categories during specific time periods.

Code of Conduct

I 192311230 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Ramesh

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Marks Evaluation:

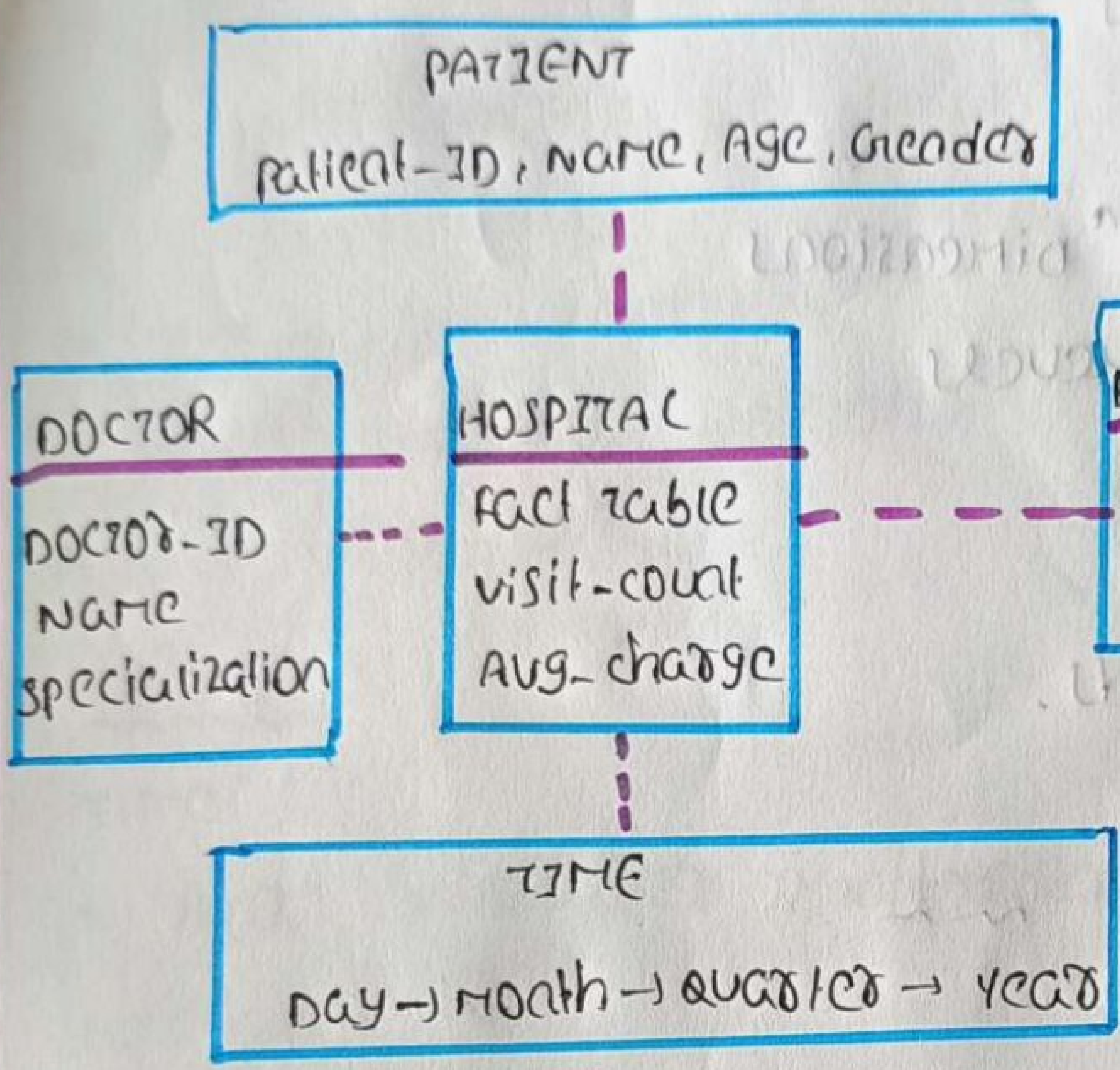
Question No.	Understanding of Concepts (25%)	Explanation (25%)	Application (20%)	Organization (15%)	Accuracy (10%)	Clarity (5%)	Total Marks
Q1 (20 Marks)							
Q2 (20 Marks)							
Q3 (20 Marks)							
Q4 (20 Marks)							
Q5 (20 Marks							
					Overall Total (100)		

Faculty In-charge	Course Coordinator	Head of Department
Signature: ______________	Signature: ______________	Signature: ______________
Date: ______________	Date: ______________	Date: ______________

① City Hospital Data warehouse

Sol-

ⓐ Snowflake schema



ⓑ OLAP operations

Start from Base cuboid
(Patient, Doctor, Department, Time)

Perform :-

* Roll-up time → year
* Roll-up patient → All patients
* Roll-up Doctor → All Doctors
* Keep Department level
* Display average consultation fee department → wise for each year.

final cuboid

(All patient, All doctor, Department, year)

(c) number of cuboids

FORMULA :-

(number of levels)^ dimension

Each dimension = 4 levels

dimension = 4

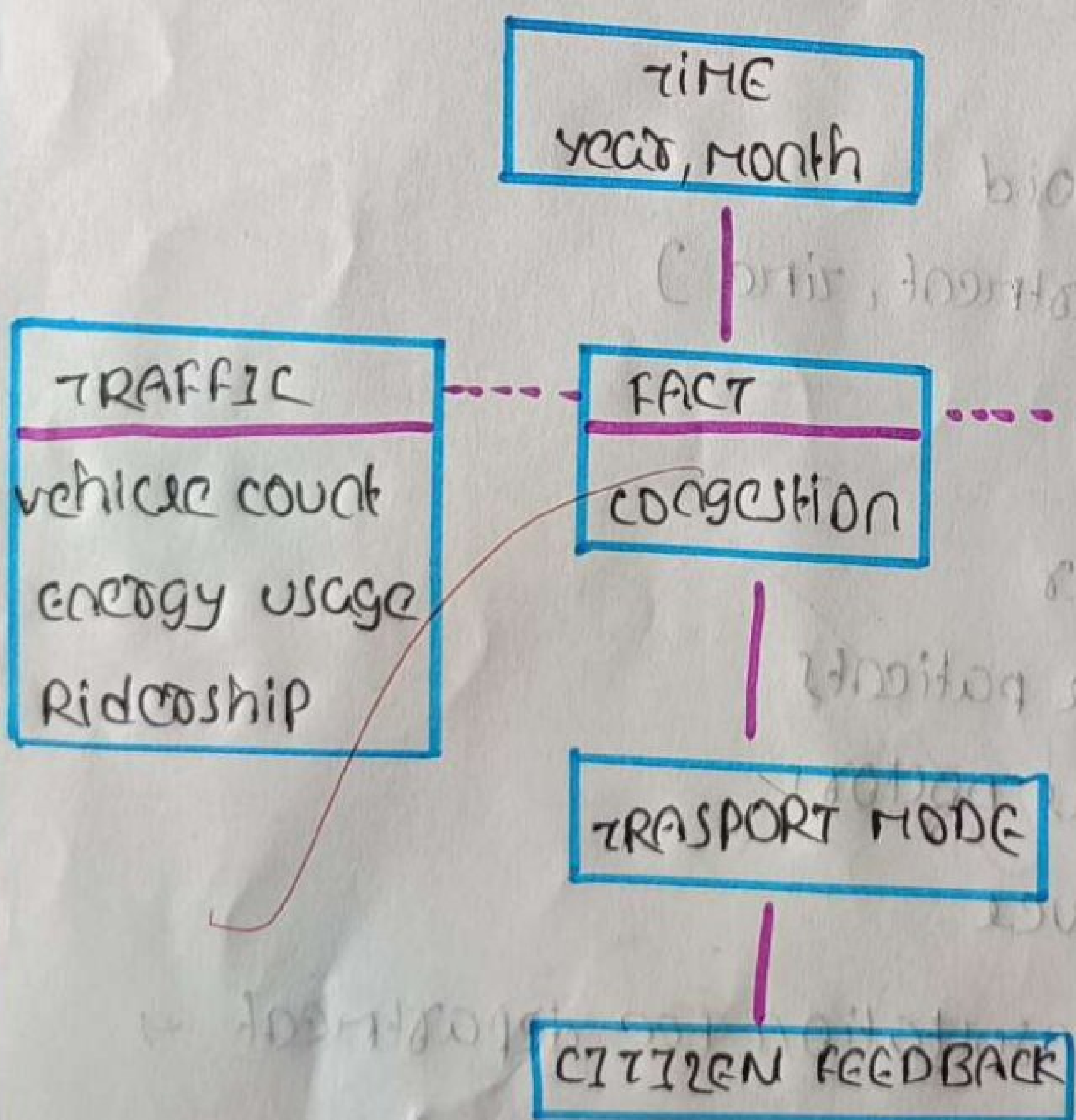
cuboids = 4x4x4x4

= 256 cuboids.

(2) sol-

smart city data warehouse

star schema



Fact table

smart-city-fact

measures

* vehicle count
* congestion level
* energy consumption
* public transport Ridership
* punctuality %

Dimensions

Time

• year • quarter • month • Day

location :-

• city • zone • Area

TRaffic :-

• Road • sensor ID

TRansport :-

• BUS • metro • TRain

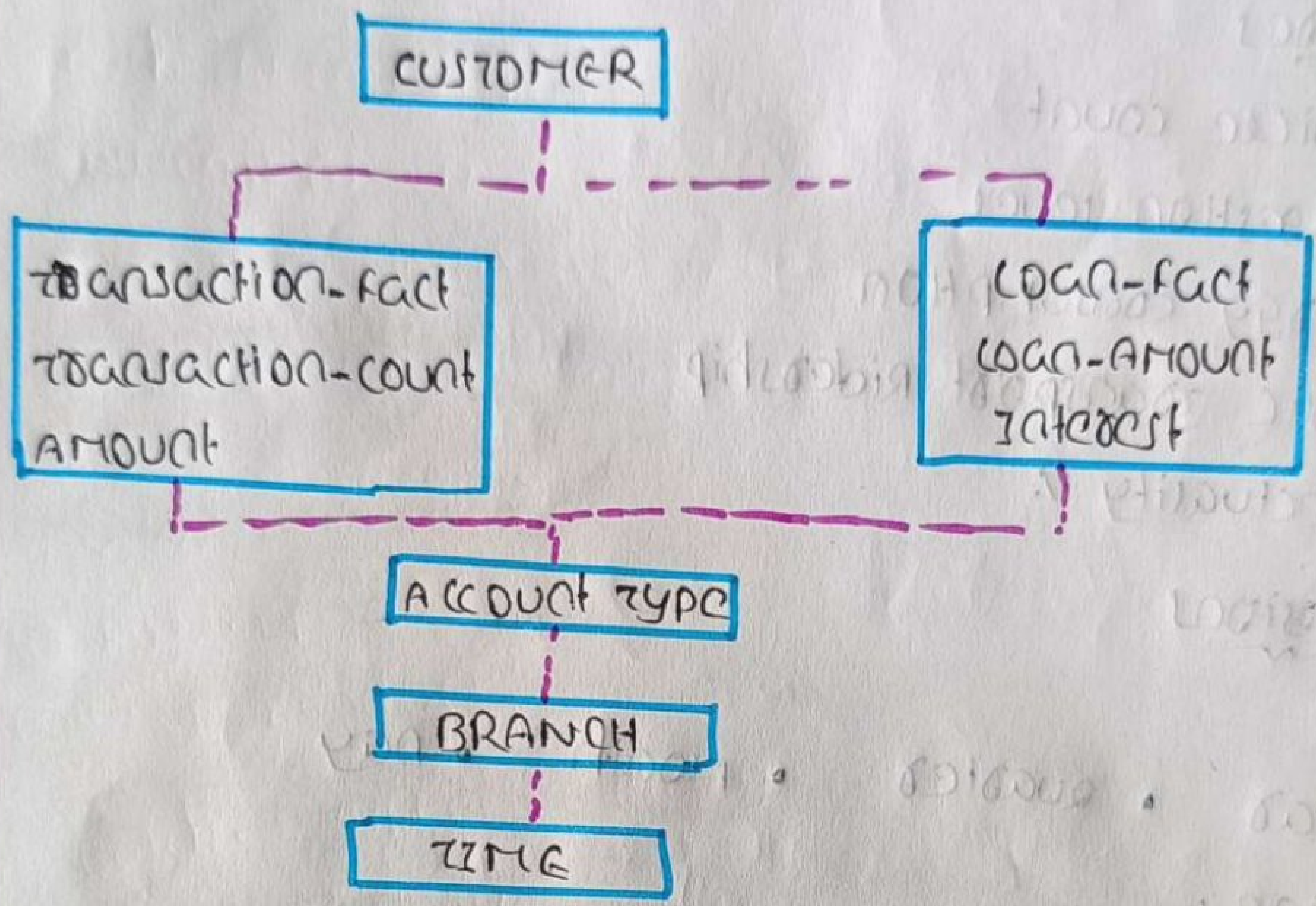
citizen :-

• feedback type
• complaint category

③ Banking Data warehouse

Sol-

ⓐ Galaxy schema



ⓑ OLAP operations

Start with

(customer, Branch, Time, Account-type)

Perform :-

* Roll up Time → year
* Roll up customer → All
* Roll up Account type → All
* keep Branch level

Result :-

[All customer, Branch, year, all account]

(c) total cuboids

each dimension has 4 levels

Dimensions = 4

$\Rightarrow 4 \times 4 \times 4 \times 4$

Answer = 256 cuboids

(4) Manufacturing company

Sol- (a) Roll-up

Roll - up product

Batch → type → category

Roll up factory

Line → Department → Factory

Result :-

Average defect rate by product category for the current year across all factories

(b) Drill-down

Factory

$\boxed{\text{Factory}}$

↓

$\boxed{\text{Department}}$

↓

$\boxed{\text{Production line}}$

product :-

$\boxed{\text{category}}$
↓
$\boxed{\text{type}}$

Analyze defect rate department-wise and product type wise

© slice

~~Ste~~ select

supplier Rating ≥ 4

Display only production units across all regions

Ⓓ Dice

choose

→ selected product categories

→ selected factory locations

Analyze

→ Defect Rate

→ supplier Reliability %

~~product :- category → type~~

3) Retail chain Data warehouse

sol-

(a) Roll-up

product

SKU → subcategory → category

store

store ID → city → Region

summarize inventory turnover by product category for the last quarter.

(b) Drill-down

Filter

electronics

then

category
↓
subcategory
↓
SKU

Analyze sales in a specific region

(c) Slice

select

Gender = Male

membership = Bronze

Display total sales revenue across all stores

(1) Dice

Select

- specific cities
- selected product categories
- particular time period

Analyze

- sales revenue
- units sold